Module 2
AWS Storage and Content Delivery Services

Module 2 Layout

- Amazon Simple Storage Service (S3)
- Amazon Elastic Block Store (EBS)
- Amazon CloudFront (CF)

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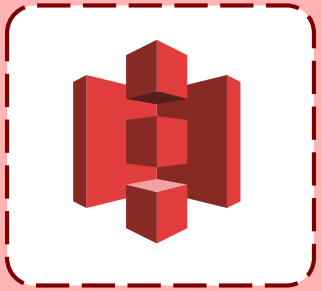
Amazon Simple Storage Service (S3)



Amazon S3

- Storage for the Internet
- Natively online, HTTP access
- Storage that allows you to store and retrieve **any amount of data**, any time, from anywhere on the web
- **Highly scalable**, reliable, fast and durable

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Amazon S3 Facts

- Can store an **unlimited number of objects** in a bucket
- Objects can be **up to 5 TB**; no bucket size limit
- Designed for **99.999999999%** durability and **99.99%** availability of objects over a given year
- Can use **HTTP/S** endpoints to store and retrieve any amount of data, at any time, from anywhere on the web
- Is highly scalable, reliable, fast, and inexpensive
- Can use optional server-side **encryption** using AWS or customer-managed provided client-side encryption
- Auditing is provided by access logs
- Provides standards-based **REST** and SOAP interfaces

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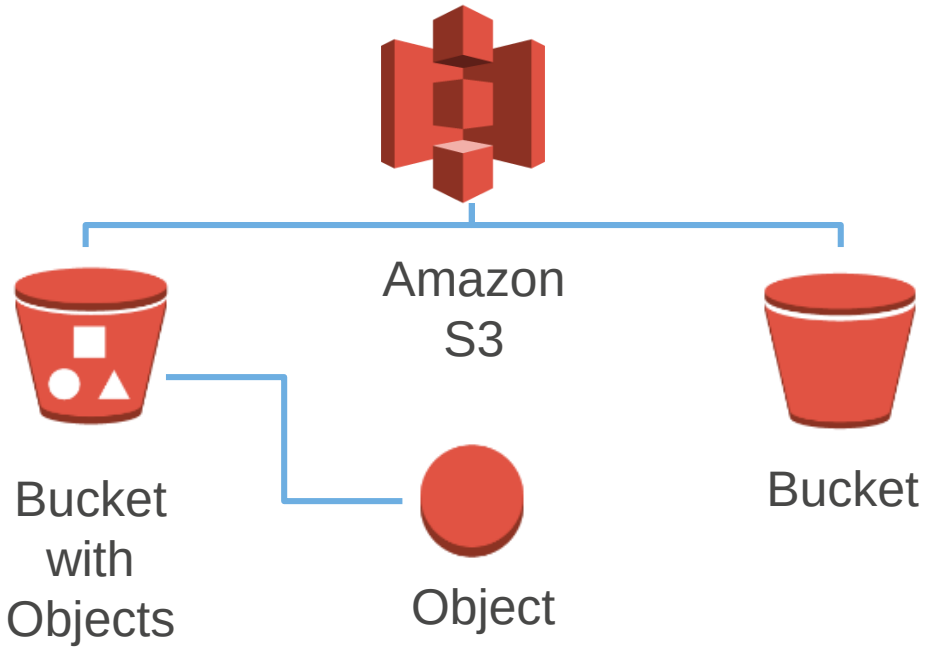
Amazon S3 Concepts



- Amazon S3 stores data as objects within **buckets**
- An object is composed of a file and optionally any **metadata** that describes that file
- You can have **up to 100 buckets** in each account
- You can **control access** to the bucket and its objects

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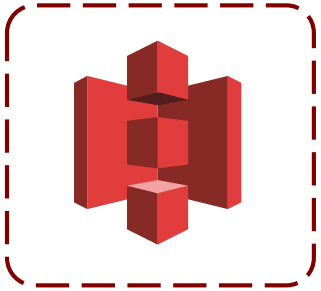
Amazon S3 Buckets



- Organize the Amazon S3 namespace at the highest level.
- Identify the account responsible for storage and data transfer charges.
- Play a role in access control.
- Serve as the unit of aggregation for usage reporting.
- Have globally unique bucket names, regardless of the AWS region in which they were created.

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Amazon S3 Region Considerations

- Amazon S3 creates a bucket in the region you select.
- You can choose a region to:
  - Optimize latency
  - Minimize costs
  - Address regulatory requirements
- Objects stored in a region never leave the region unless you explicitly transfer them to another region.

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Amazon S3 Objects

- Objects are the fundamental entities stored in Amazon S3.
- When using the console, you can think of them as files.
- **Objects consist of data and metadata**. The data portion is opaque to Amazon S3. The metadata is a set of name-value pairs that describe the object.
    - Default metadata such as the date last modified
    - Standard HTTP metadata such as Content-Type
    - Custom metadata at the time the object is stored
    - A key that uniquely identifies as object within its bucket

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Object Keys

An object key is the unique identifier for an object in a bucket.

http://doc.s3.amazonaws.com/2006-03-01/AmazonS3.html

Bucket

Object/Key

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Amazon S3 Security

- You can **control access** to buckets and objects with:
  - Access Control Lists (ACLs)
  - Bucket policies
  - Identity and Access Management (IAM) policies
- You can upload or download data to Amazon S3 via **SSL** encrypted endpoints.
- You can **encrypt data** using AWS SDKs.

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Amazon S3 Versioning

- Protects from **accidental overwrites and deletes** with no performance penalty.
- Generates a **new version with every upload**.
- Allows easily retrieval of deleted objects or **roll back** to previous versions.
- Three states of an Amazon S3 bucket
  - Un-versioned (default)
  - Versioning-enabled
  - Versioning-suspended

Versioning Enabled

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Amazon S3 Object Lifecycle

Lifecycle management defines how Amazon S3 manages objects during their lifetime. Some objects that you store in an Amazon S3 bucket might have a well-defined lifecycle:

- Log files
- Archive documents
- Digital media archives
- Financial and healthcare records
- Raw genomics sequence data
- Long-term database backups
- Data that must be retained for regulatory compliance

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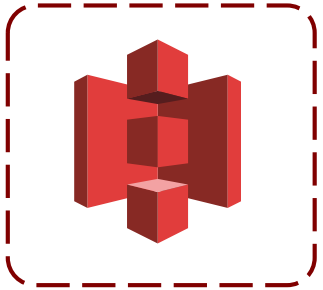
Common Use Scenarios



- Storage and backup
- Application file hosting
- Media hosting
- Software delivery
- Store AMIs and snapshots

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Landsat on AWS

We have committed to make up to 1 petabyte of Landsat imagery readily available as objects on Amazon S3.

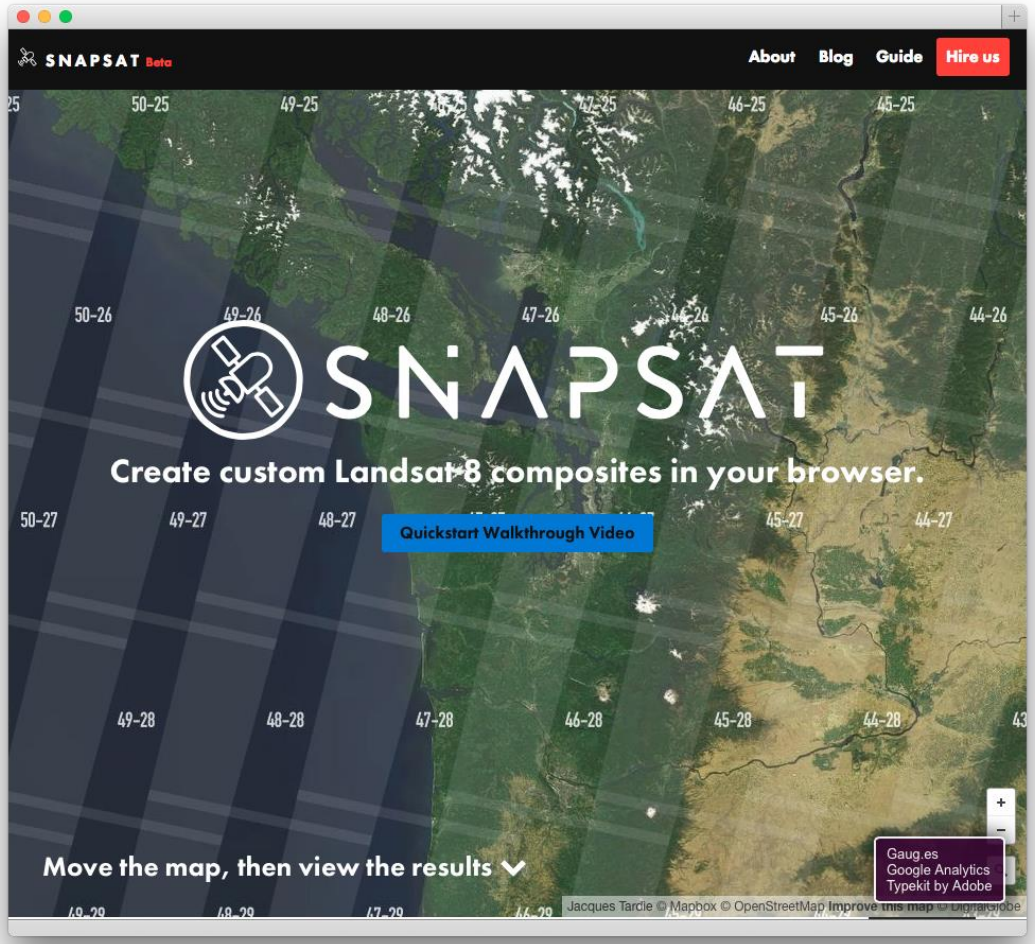
All Landsat 8 scenes from 2015 are available, along with a selection of cloud-free scenes from 2013 and 2014. All new Landsat 8 scenes are made available each day (~700 per day), often within hours of production.

Now anyone can work with Landsat data with no up-front investment of time or capital expense.

Snapsat

A team of five novice programmers used Landsat on AWS to build a web app called Snapsat that creates Landsat data visualizations in seconds.

Snapsat was built during the team's 8-week training program at Code Fellows. They launched it just months after learning to write code.



http://snapsat.org

Amazon S3 Pricing

- Pay only for what you use
- No minimum fee
- Prices based on location of your Amazon S3 bucket
- Estimate monthly bill using the **AWS Simple Monthly Calculator**
- Pricing is available as:
  - Storage Pricing
  - Request Pricing
  - Data Transfer Pricing: data transferred out of Amazon S3

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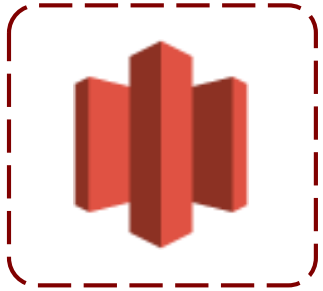
Amazon Glacier



- Long term low-cost archiving service
- Optimal for infrequently accessed data
- Designed for 99.999999999% durability
- Three to five hours' retrieval time
- Less than $0.01 per GB/month (depending on region)

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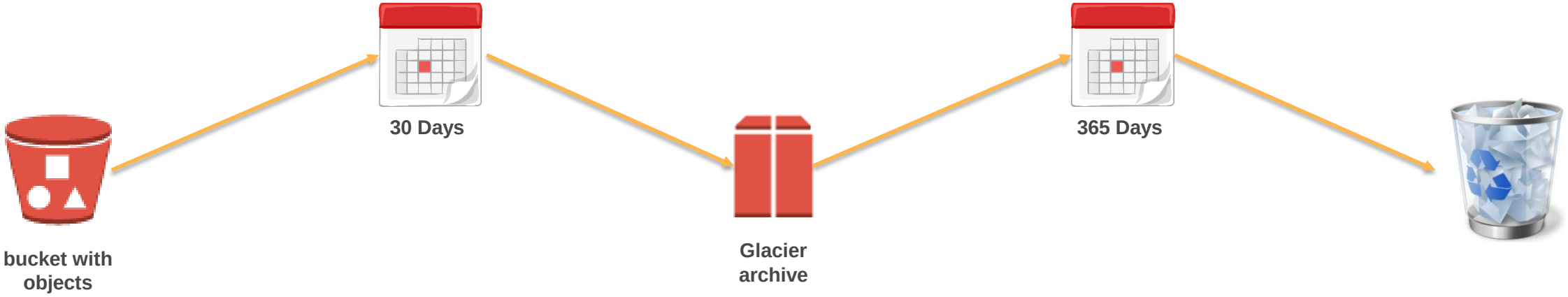
Amazon S3 + Amazon Glacier

S3 Lifecycle policies allow you to delete or move objects based on age and set rules per S3 bucket.



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Amazon S3 Storage Classes

Storage Class	Durability	Availability	Other Considerations
Amazon S3 Standard	99.999999999%	99.99%	
Amazon S3 Standard - Infrequent Access (IA)	99.999999999%	99.9%	• Retrieval fee associated with objects • Most suitable for infrequently accessed data
Glacier	99.999999999%	99.99% (once restored)	• Not available for real-time access • Must restore objects before you can access them • Restoring objects can take 3-5 hours

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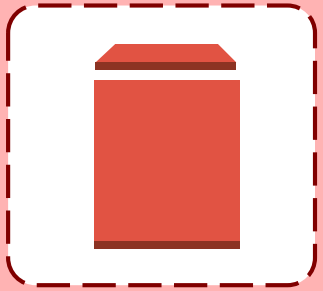
Instructor Demo
Amazon S3

Amazon Elastic Block Store (EBS)



Amazon EBS

- **Persistent block level storage** volumes offer consistent and low-latency performance.
- Stored data is automatically replicated within its Availability Zone.
- Snapshots are stored durably in Amazon S3.

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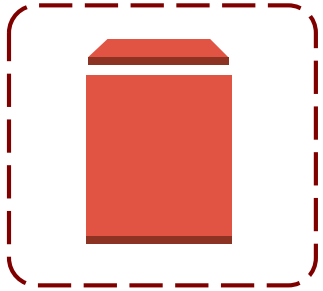
Amazon EBS Lifecycle



Vast amounts of unused space

Create

Call CreateVolume
1 GiB to 16 TiB

Attach

Call AttachVolume to affiliate with one Amazon EC2 instance

Attached and In Use

- Format from Amazon EC2 instance OS
- Mount formatted drive

CreateSnapshot

Snapshot to Amazon S3

Detach

Call DetachVolume

Deleted

Call DeleteVolume

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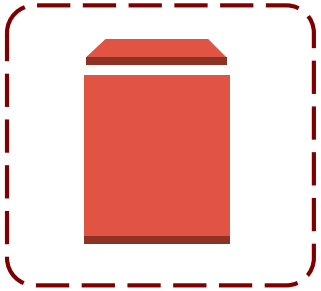
Amazon EBS Volume Types



- SSD-backed volumes are
  - Optimized for **transactional** workloads that involve **frequent read/write** operations with **small I/O** size.
  - Dominant in **IOPS** performance.
- HDD-backed volumes are
  - Optimized for **large streaming** workloads.
  - Dominant in **throughput** (measured in MiB/s).

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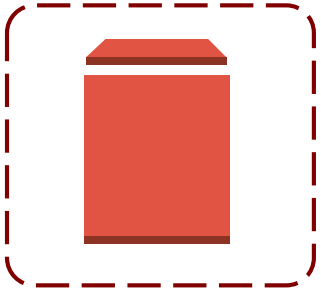
Amazon EBS Volume Types



	SSD		HDD	
Volume Type	**General Purpose SSD (gp2)**	**Provisioned IOPS SSD (io1)**	**Throughput Optimized HDD (st1)**	**Cold HDD (sc1)**
Description	Balances price and performance for a wide variety of **transactional** loads.	**Highest-performance** SSD volume designed for **mission-critical** applications.	**Low-cost** HDD designed for **frequently accessed**, **throughput-intensive** workloads.	**Lowest cost** HDD designed for less frequently accessed workloads.
Volume Sizes	1 GiB – 16 TiB	4 GiB – 16 TiB	500 GiB – 16 TiB	500 GiB – 16 TiB
Dominant Performance Attribute	IOPS	IOPS	MiB/s	MiB/s

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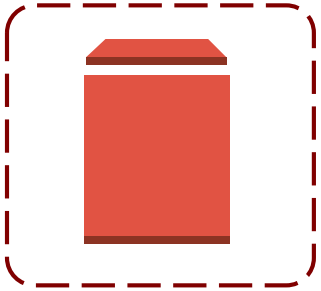
Amazon EBS Facts



- EBS is recommended when data must be **quickly accessible** and requires **long-term persistence**.
- You can launch your EBS volumes as **encrypted** volumes – data stored at rest on the volume, disk I/O, and snapshots created from the volume are all encrypted.
- You can create **point-in-time snapshots** of EBS volumes, which are persisted to Amazon S3.

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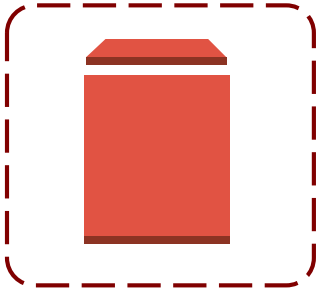
Amazon EBS Use Cases



- **OS:** Use for boot/root volume, secondary volumes
- **Databases:** Scales with your performance needs
- **Enterprise applications:** Provides reliable block storage to run mission-critical applications
- **Business continuity:** Minimize data loss and recovery time by regularly backing up using EBS Snapshots
- **Applications:** Install and persist any application

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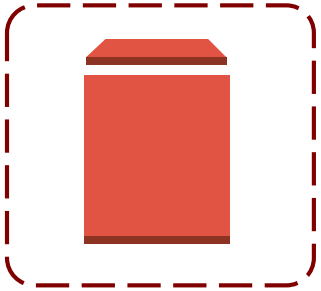
Amazon EBS Pricing



Pay for what you provision:

- Pricing based on region
- Review Pricing Calculator online
- Pricing is available as:
  - Storage
  - IOPS

** Check Amazon EBS Pricing page for current pricing for all regions.*

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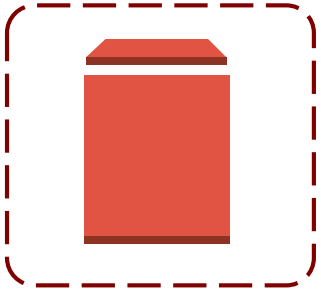
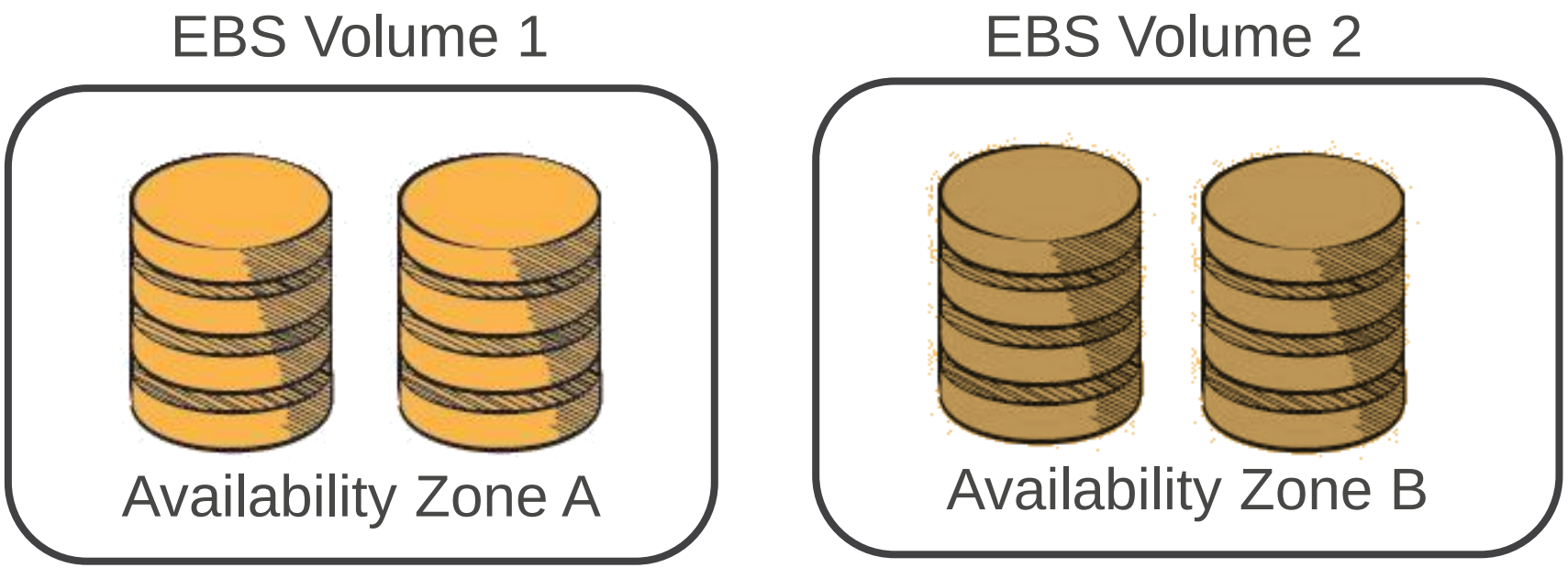
Amazon EBS Scope



Amazon EBS volumes are in a single Availability Zone



Volume data is replicated across multiple servers in an Availability Zone.

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Amazon EBS and Amazon S3

	Amazon EBS	Amazon S3
Paradigm	Block storage with file system	Object store
Performance	Very fast	Fast
Redundancy	Across multiple servers in an Availability Zone	Across multiple facilities in a Region
Security	EBS Encryption – Data volumes and Snapshots	Encryption
Access from the Internet?	No (1)	Yes (2)
Typical use case	It is a disk drive	Online storage

(1) Accessible from the Internet if mounted to server and set up as FTP, etc.
(2) Only with proper credentials, unless ACLs are world-readable

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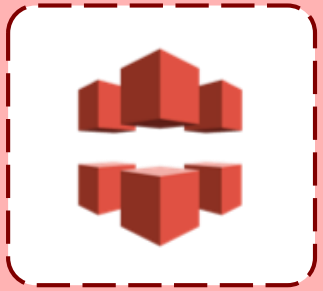
Amazon CloudFront

Amazon CloudFront



Amazon CloudFront

- Easy and cost effective way to **distribute content** to end users
- **Low latency, high data transfer speeds**
- Deliver your entire website, including static, dynamic, and streaming content using a global network of edge locations

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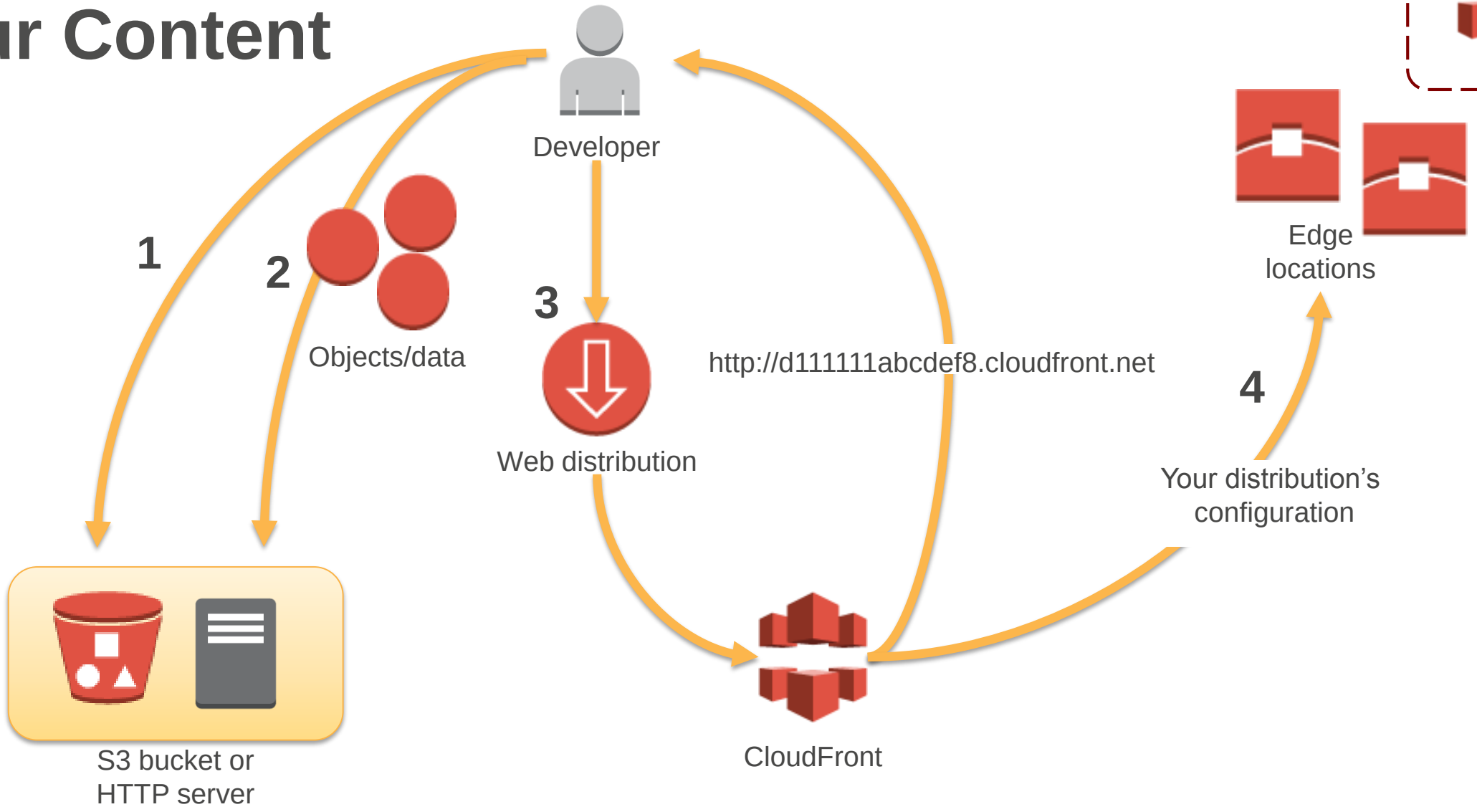
How You Configure CloudFront to Deliver Your Content



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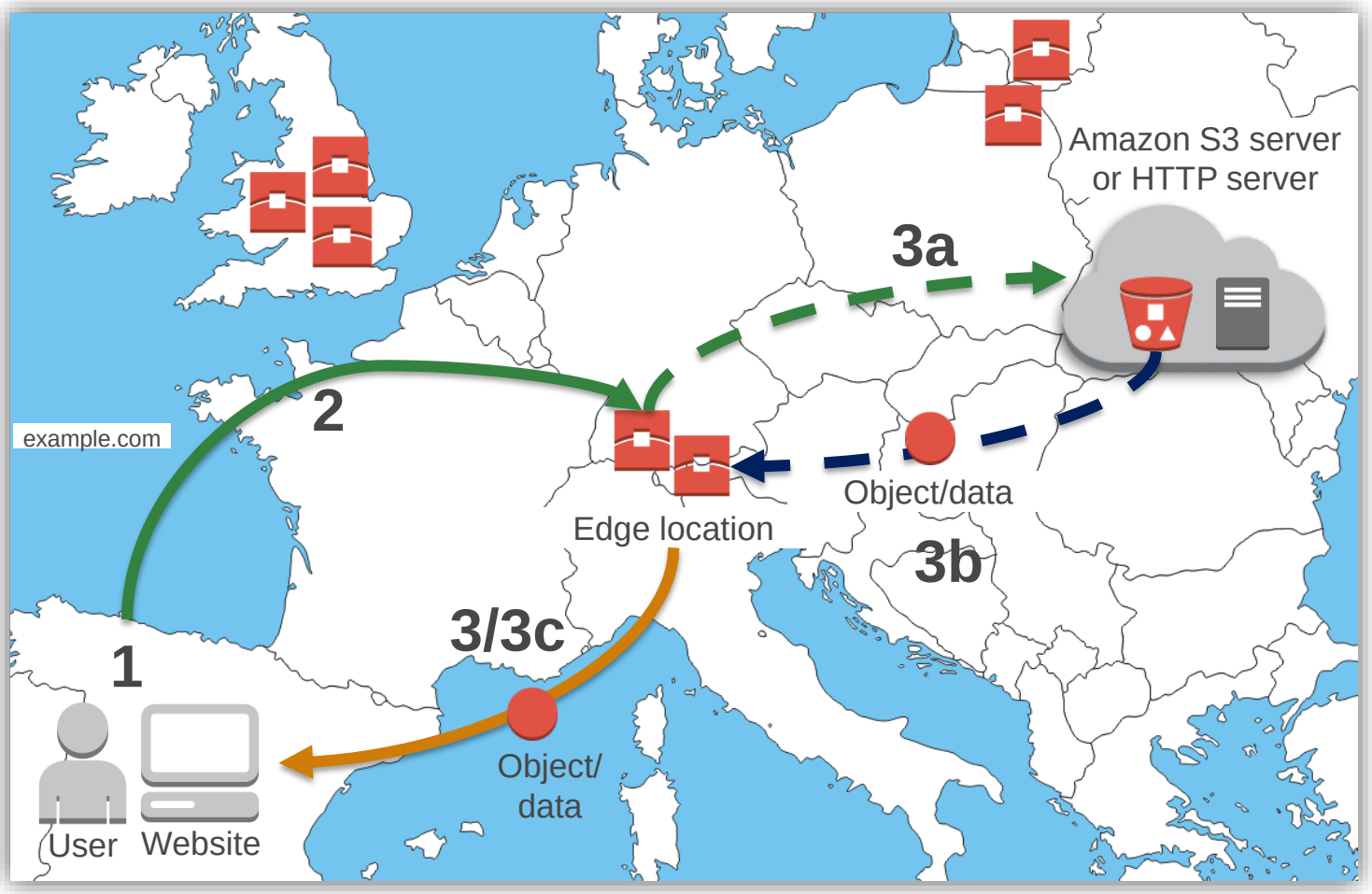
How CloudFront Delivers Content to Your Users



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Q & A

Lunch Time !